CHRISTIAN OMAR PAYÁN TORRÓNTEGUI

669 100 3234 • omarpayant@gmail.com • linkedin.com/in/omarpayant/ • github.com/opyntorr

SUMMARY

Robotics & Digital Systems Engineering student focused on product design, advanced manufacturing, and hardware reliability. Proven experience in 3D modeling, additive manufacturing, and mechanical interfaces. Seeking to leverage mechanical design to build practical, robust solutions for industrial applications.

EDUCATION

B.S. Robotics and Digital Systems Graduating Expected December 2026
Tecnológico de Monterrey
Zapopan, Jalisco
Relevant coursework: Jewelry Concentration: Advanced 3D modeling and product design using Rhino 3D

TECHNICAL SKILLS

Programming & Languages: Python, C++, Spanish (Native), English (B2)
Robotics & Simulation: ROS 2, Gazebo, URDF
Design & Electronics: SOLIDWORKS, Fusion 360, Rhino 3D, Eagle
Office & Productivity: Microsoft Office (Excel, Word, PowerPoint, Outlook — 100-hour certification)
Tools & Methodologies: Git, Agile, Scrum, Advanced FDM 3D Printing, Linux (Ubuntu), Docker

PROFESSIONAL EXPERIENCE

Industrial Mounts & Robotic Support Interfaces, Zapopan, Jalisco: Mechanical & Product Designer Feb-Jun 2026

- Engineered mechanical interfaces for VR controllers (Meta Quest 3), developing a 3-DOF system that reduced fabrication costs by 95% vs commercial alternatives
- Designed and prototyped a custom mini PC enclosure for robust deployment in industrial environments, utilizing 3D-printed fiberglass-reinforced PETG for its ideal thermal and mechanical properties

ACADEMIC & TECHNICAL PROJECTS

Open-Source Gastrointestinal Medical Phantom Ago-Dic 2025 – Feb-Jun 2026
Agile methodologies, Advanced FDM

- Engineered phantom via advanced FDM 3D printing (TPU/PETG), reducing prototyping costs by 80% vs commercial alternatives
- Managed development using Agile methodologies to iterate rapidly on anatomical accuracy and material constraints

DigiMix Ago-Dic 2024
Embedded Systems, Scrum

- Designed custom PCB and 3D-printed casing; engineered hardware using dual-core STM32 and ESP32 (WebSockets)
- Applied Scrum to manage lifecycle, integrating hardware and software architectures with zero logic faults